MINI PROJECT REPORT
ON
"SMART EV CHARGER MONITORING AND PROTECTION SYSTEM"

Submitted By

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Preamble

The rapid growth of electric vehicles (EVs) has created a significant demand for efficient, reliable, and intelligent charging infrastructure. As the world transitions toward sustainable energy solutions, it becomes essential to ensure that EV charging systems are not only accessible but also safe and energy-efficient. Conventional charging systems often lack real-time monitoring and protection mechanisms, which can lead to issues such as overcurrent, overheating, and inefficient energy usage. Therefore, there is a growing need to develop smart charging systems that integrate advanced sensing, control, and communication technologies.

This project presents the design and development of a Smart EV Charger Monitoring and Protection System using modern embedded technology. The system incorporates sensors such as current, voltage, and temperature sensors to continuously monitor the charging conditions. An ESP32 microcontroller acts as the core processing unit, enabling real-time data acquisition and wireless communication. With the integration of IoT platforms, users can remotely monitor charging parameters and control the system through a mobile application, ensuring enhanced convenience and safety.

The primary objective of this project is to improve the reliability and safety of EV charging by implementing automatic protection features and intelligent monitoring. The system is designed to detect abnormal conditions such as overcurrent or overheating and take necessary actions like disconnecting the load using a relay module. This not only protects the charging equipment but also extends the lifespan of the battery and ensures user safety. Overall, the project aims to contribute to the advancement of smart and sustainable EV charging solutions.

CERTIFICATE

This is to certify that

Mr. Samarth Phakatkar

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Branch E&TC Engineering has successfully completed the work associated with **Mini Project (304200**) titled as "**SMART EV CHARGER MONITORING AND PROTECTION SYSTEM**" and has the work book associated under my supervision, in the partial fulfillment of Third Year Bachelor of Engineering (Choice Based Credit System) (2019 course) of Savitribai Phule Pune University.

Date:

Place: ICEM, Pune

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External Examiner

ACKNOWLEDGEMENT

With immense pleasure, we are presenting this Mini Project report as a part of the curriculum of Third Year Bachelor of Engineering (B.E.) in Electronics and Telecommunication Engineering (E&TC). We would like to express our sincere gratitude to all those who supported us throughout the development of this project.

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We also express our gratitude to everyone who directly or indirectly contributed to the successful completion of this project.

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ABSTRACT

The increasing adoption of electric vehicles (EVs) has highlighted the need for advanced and reliable charging systems. Traditional EV chargers often lack real-time monitoring and protection features, which can lead to safety risks such as overcurrent, voltage fluctuations, and overheating. To address these challenges, this project proposes a Smart EV Charger Monitoring and Protection System that ensures safe, efficient, and intelligent charging operations using embedded and IoT technologies.

The system is built around an ESP32 microcontroller, which collects real-time data from sensors including the ACS712 current sensor, ZMPT101B voltage sensor, and a temperature sensor. These parameters are continuously monitored to analyze the charging conditions and ensure efficient operation of the EV charger.

The integration of an IoT platform enables remote monitoring and control through a mobile application, allowing users to observe live data and operate the charger conveniently. A relay module is used to automatically disconnect the power supply in case of abnormal conditions, thereby preventing potential damage and enhancing overall system safety

The proposed system offers a cost-effective and scalable solution for modern EV charging infrastructure. It improves reliability by incorporating protection mechanisms and enhances user experience through smart monitoring features. This project demonstrates how the integration of IoT and sensor-based automation can significantly contribute to the development of safe and efficient EV charging systems, supporting the transition toward sustainable transportation.

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INTRODUCTION

The transportation sector is undergoing a major transformation with the increasing adoption of electric vehicles (EVs) as a sustainable alternative to conventional fuel-based vehicles. EVs help reduce carbon emissions, minimize dependence on fossil fuels, and promote environmentally friendly mobility. However, the rapid growth in EV usage has created a strong demand for efficient, reliable, and safe charging infrastructure. Ensuring proper charging mechanisms is crucial for maintaining battery health, user safety, and overall system performance.

Conventional EV charging systems primarily focus on delivering power to the vehicle but often lack advanced monitoring and protection features. Issues such as overcurrent, voltage instability, and overheating can lead to equipment damage, reduced battery life, and potential safety hazards. Therefore, it is essential to develop smart charging solutions that can continuously monitor key electrical parameters and take corrective actions in real time to prevent failures and ensure efficient operation.

In this project, a Smart EV Charger Monitoring and Protection System is proposed using embedded systems and IoT technology. The system utilizes sensors such as ACS712 for current measurement, ZMPT101B for voltage sensing, and a temperature sensor to track thermal conditions. An ESP32 microcontroller acts as the central processing unit, collecting sensor data and enabling wireless communication. The integration of IoT allows users to remotely monitor charging parameters and control the system through a mobile application, providing convenience and flexibility.

The proposed system enhances the safety and reliability of EV charging by incorporating automatic protection mechanisms. In case of abnormal conditions such as excessive current or high temperature, the system triggers a relay to disconnect the power supply, preventing damage to the charger and the vehicle. This approach not only ensures safe operation but also contributes to energy efficiency and long-term sustainability. The project demonstrates the potential of combining IoT and sensor-based technologies to develop intelligent EV charging solutions for modern applications.

PROBLEM STATEMENT

The rapid growth in electric vehicle (EV) adoption has increased the demand for reliable and intelligent charging infrastructure. Conventional EV charging systems mainly focus on delivering power to the vehicle but lack real-time monitoring and protection features. This can lead to inefficient charging, equipment damage, and potential safety hazards, affecting both user experience and battery lifespan.

Furthermore, issues such as overcurrent, voltage fluctuations, and overheating are common during charging due to varying load conditions and environmental factors. Most existing systems do not provide remote access or automatic fault detection, requiring manual intervention and increasing the risk of accidents.

Therefore, there is a need for a smart EV charging system that ensures continuous monitoring and automatic protection for safe and efficient operation.

Hence, there is a critical need to develop a smart, safe, and cost-effective EV charging system that can:

- Provide Real-Time Monitoring: Continuously track parameters like current, voltage, and temperature.
- Ensure Protection Against Faults: Detect overcurrent, overvoltage, and overheating conditions.
- Enable Remote Monitoring and Control: Allow users to monitor and control the system through IoT platforms.
- Reduce Risk of Equipment Damage: Automatically disconnect power during abnormal conditions.
- Improve System Reliability and Efficiency: Ensure stable and optimized charging performance.

The objective is to design and implement an automated EV charging system using advanced sensors and IoT technology to enhance safety, improve efficiency, and provide a user-friendly smart charging solution.

LITERATURE SURVEY

[1] R. Gupta et al. (2020)

R. Gupta et al. (2020) proposed an IoT-based smart energy monitoring system using embedded controllers and wireless communication. The system enables real-time monitoring of voltage and current parameters and provides remote access through a mobile interface. By integrating sensors with microcontrollers, the study demonstrates improved efficiency in energy management and fault detection. The proposed system highlights the importance of real-time monitoring for electrical safety and energy optimization.

[2] S. Patel & M. Shah (2021)

S. Patel and M. Shah (2021) developed a smart EV charging station with integrated protection mechanisms. The system uses voltage and current sensors to monitor charging conditions and automatically disconnects the supply during abnormal situations such as overvoltage and overcurrent. Their approach enhances the safety of EV charging infrastructure and reduces the risk of damage to the vehicle battery and charging unit.

[3] A. Kumar et al. (2022)

A. Kumar et al. (2022) designed an ESP32-based real-time monitoring system for electrical applications. The system collects sensor data such as temperature, voltage, and current, and processes it using embedded algorithms. The study emphasizes the role of microcontrollers in smart monitoring systems and demonstrates how real-time data acquisition improves system reliability and performance.

[4] J. Lee et al. (2023)

J. Lee et al. (2023) presented a temperature-based protection system for electrical devices using digital sensors. The system continuously monitors thermal conditions and triggers a shutdown mechanism when the temperature exceeds safe limits. Their work highlights the importance of thermal monitoring in preventing equipment failure and ensuring safe operation in high-load electrical systems.

[5] K. Verma et al. (2022)

K. Verma et al. (2022) developed a smart energy metering system using IoT technology. The system measures power consumption and provides real-time data to users for analysis and billing purposes. The research demonstrates how integrating monitoring and analytics can improve energy efficiency and enable intelligent decision-making in electrical systems.

[6] T. Nguyen et al. (2024)

T. Nguyen et al. (2024) proposed an advanced EV charging infrastructure with integrated fault detection and monitoring capabilities. The system combines multiple sensors and control mechanisms to ensure safe and efficient charging. The study shows that combining monitoring, protection, and automation significantly enhances the reliability of EV charging systems.

[7] R. Gupta et al., 2020

This paper presents an IoT-based system for monitoring and controlling household electrical appliances. It focuses on real-time energy consumption tracking using sensors and microcontrollers. The system allows users to remotely monitor energy usage and optimize power consumption. The concept of real-time monitoring and control is relevant to EV charging systems, as it improves efficiency and user awareness.

[8] T. Nguyen et al., Journal of Energy Systems, 2024

This study discusses the development of an advanced EV charging system with built-in fault detection mechanisms. It emphasizes monitoring electrical parameters such as voltage and current to identify abnormal conditions. The system enhances safety by detecting faults and preventing damage to the charger and vehicle. This work highlights the importance of intelligent monitoring and protection in modern EV charging infrastructure.

MOTIVATION

The increasing shift toward electric vehicles (EVs) as a sustainable mode of transportation is one of the major motivations behind this project. With rising environmental concerns and the need to reduce carbon emissions, EVs are becoming more popular worldwide. However, the effectiveness of EV adoption largely depends on the availability of safe and efficient charging systems, which motivates the development of improved charging solutions.

Another key motivation is the lack of intelligent monitoring and protection in conventional EV chargers. Many existing systems do not provide real-time information about critical parameters such as current, voltage, and temperature. This can lead to unsafe charging conditions, reduced battery life, and possible equipment damage. Addressing these limitations by implementing smart monitoring features is an important driving factor for this project.

The advancement of Internet of Things (IoT) technology also plays a significant role in motivating this work. IoT enables remote monitoring and control of devices, making systems more user-friendly and efficient. By integrating IoT with EV charging, users can conveniently track charging status and control operations through a mobile application, enhancing overall user experience.

Finally, ensuring safety and reliability in EV charging systems is a major concern. Fault conditions such as overcurrent and overheating can lead to serious risks if not handled properly. This project is motivated by the need to design a system that not only monitors these parameters but also provides automatic protection using relay-based control. The aim is to develop a smart, efficient, and secure EV charging solution suitable for modern applications.

OBJECTIVES

• To develop a smart EV charging system

The primary objective is to design a system that ensures safe and efficient charging of electric vehicles. By integrating sensors and a microcontroller-based system, the charger can operate intelligently with minimal human intervention and improved reliability.

• To monitor charging parameters in real time

The system aims to continuously measure important parameters such as current, voltage, and temperature. This real-time monitoring helps in analyzing charging conditions and maintaining optimal performance of the EV battery.

• To implement automatic protection mechanisms.

One of the key objectives is to protect the system from faults such as overcurrent, overvoltage, and overheating. The system uses a relay to automatically disconnect the power supply during abnormal conditions, ensuring safety of both the charger and the vehicle.

• To enable remote monitoring and control

The project focuses on integrating IoT technology to allow users to monitor and control the charging system remotely through a mobile application. This enhances convenience and provides better control over the charging process.

• To improve system reliability and efficiency

The system is designed to ensure stable and efficient charging by minimizing energy loss and preventing damage to components. This contributes to longer battery life and better overall performance.

METHODOLOGY / PROPOSED SYSTEM

BLOCK DIAGRAM

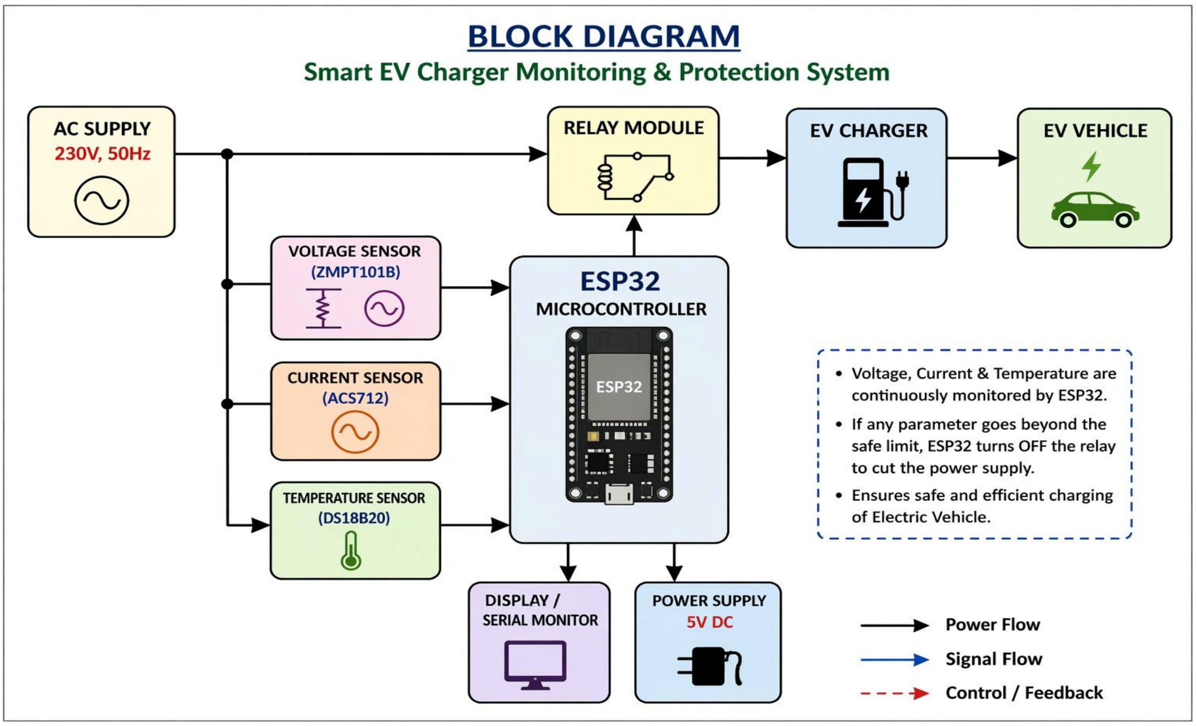


Fig 1: Block Diagram of Smart EV Charger Monitoring and Protection System

The working of the Smart EV Charger Monitoring and Protection System is carried out in a sequence of steps to ensure safe and efficient charging operation.

Step 1: Power Supply Initialization

The system is powered using a regulated power supply. The ESP32, sensors, relay module, and LCD display are initialized and made ready for operation.

Step 2: Sensor Data Acquisition

The ACS712 current sensor, ZMPT101B voltage sensor, and temperature sensor continuously measure real-time values from the EV charging system.

Step 3: Data Processing by ESP32

The ESP32 reads the analog values from sensors, converts them into meaningful electrical parameters, and processes the data for monitoring.

Step 4: Display of Parameters

The measured values such as current, voltage, and temperature are displayed on the LCD screen for real-time local monitoring.

Step 5: IoT Data Transmission

The ESP32 sends the processed data to the IoT platform (e.g., Blynk app) via Wi-Fi, enabling remote monitoring through a mobile application.

Step 6: Continuous Monitoring and Comparison

The system continuously compares sensor values with predefined safe threshold limits to check for abnormal conditions.

Step 7: Fault Detection

If any parameter exceeds the safe limit (overcurrent, overvoltage, or high temperature), the system identifies it as a fault condition.

Step 8: Relay Activation for Protection

Upon fault detection, the relay module is triggered to disconnect the power supply, protecting the EV charger and battery from damage.

Step 9: User Alert and Notification

The system sends alerts or notifications to the user via the IoT application, informing them about the fault condition.

Step 10: System Recovery / Manual Control

After the fault is resolved, the user can restart the system manually or through the mobile app, restoring normal operation.

HARDWARE REQUIREMENT

- ESP32 Microcontroller
- ACS712 Current Sensor
- ZMPT101B Voltage Sensor
- Temperature Sensor (LM35 / DHT11)
- Relay Module
- 16x2 LCD Display (I2C)
- Power Supply Unit
- Connecting Wires
- EV Charging Load (Bulb/Demo Load)

COMPONENTS SPECIFICATION

1. ESP32 Microcontroller

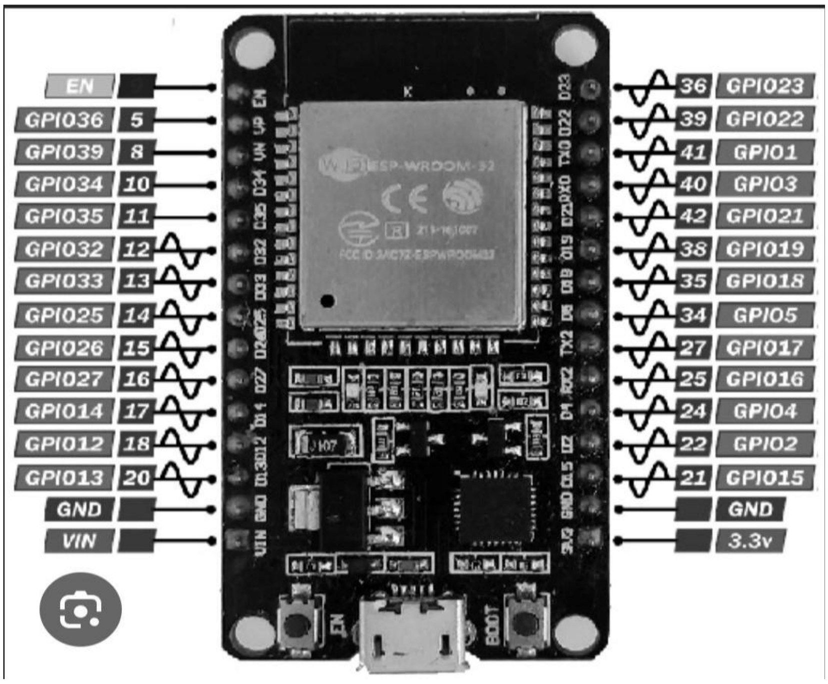


Fig 2: ESP 32 Microcontroller

Operating Voltage: 3.3V

Wi-Fi: 802.11 b/g/n

Bluetooth: v4.2 (BLE)

Clock Speed: up to 240 MHz

GPIO Pins: ~30

Features: Dual-core processor, low power consumption

2. ACS712 Current Sensor

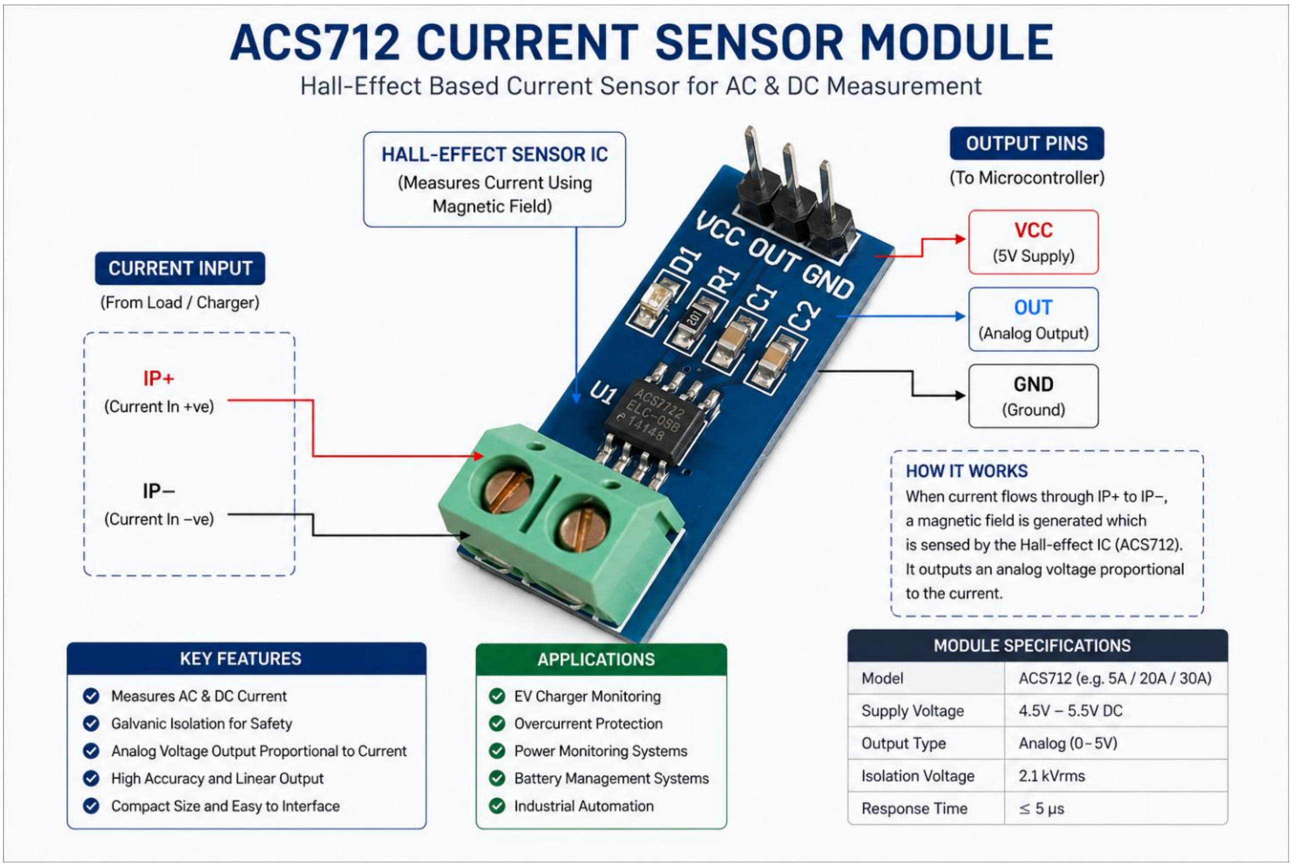


Fig 3 : ACS712 Current Sensor

Operating Voltage: 5V

Current Range: ±5A / ±20A / ±30A (depends on model)

Output Type: Analog voltage

Sensitivity: 185mV/A (5A), 100mV/A (20A), 66mV/A (30A)

Output Offset: ~2.5V (at 0A)

Response Time: 5µs

Isolation Voltage: 2.1kV RMS

Features: Hall-effect sensing, electrical isolation, compact size

3. ZMPT101B Voltage Sensor

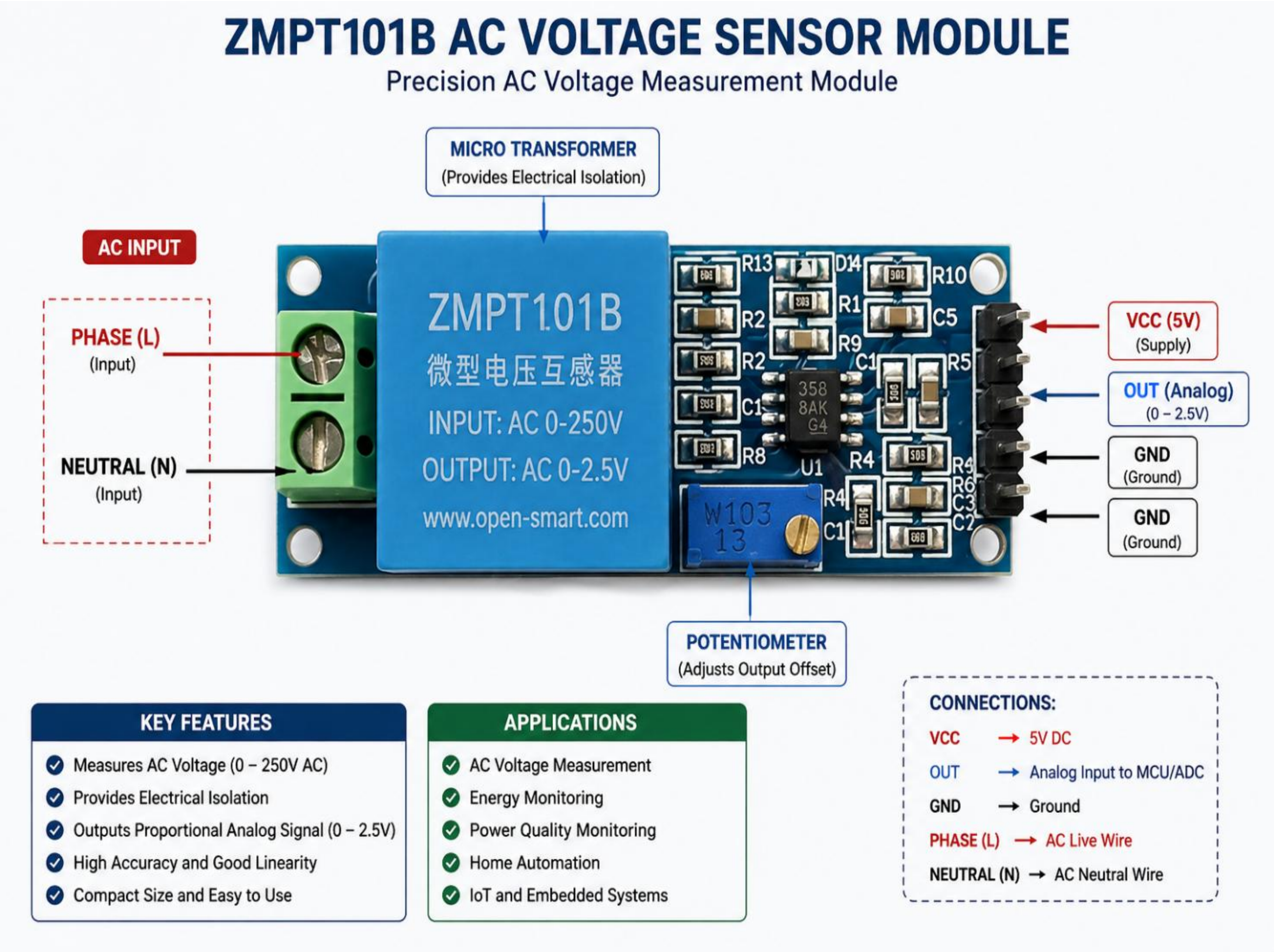


Fig 4 : ZMPT101B Voltage Sensor

Operating Voltage: 5V

Input Voltage Range: 0–250V AC

Output Type: Analog voltage

Sensitivity: Adjustable (via onboard potentiometer)

Isolation: Transformer-based electrical isolation

Frequency Range: 50Hz / 60Hz

Accuracy: High (with proper calibration)

Features: Compact size, safe AC measurement, easy interface with microcontrollers.

4. Temperature Sensor (LM35)

Fig 5 : Temperature Sensor (LM35)

Operating Voltage: 4V – 30V
Temperature Range: -55°C to 150°C
Output Type: Analog voltage
Accuracy: ±0.5°C (at 25°C)
Sensitivity: 10mV/°C
Features: Linear output, no calibration required, low self-heating

5. Relay Module

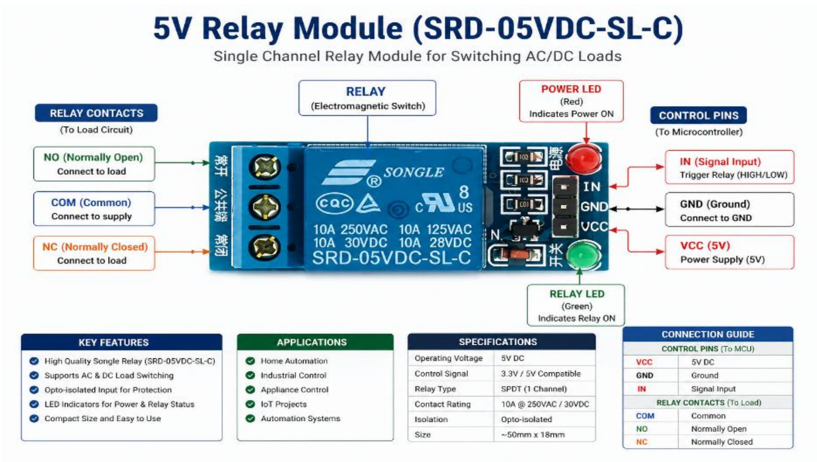


Fig 6 : Relay Module

Operating Voltage: 5V DC

Trigger Voltage: 3.3V / 5V (compatible with ESP32)

Relay Type: SPDT (Single Pole Double Throw)

Contact Rating: 10A @ 250V AC / 10A @ 30V DC

Control Signal: Digital (HIGH/LOW)

Channels: 1 Channel

Isolation: Opto-isolated (in most modules)

Features: LED indicators, safe switching of high voltage loads, easy interface.

SOFTWARE REQUIREMENT

1. **Arduino IDE**

   Arduino IDE is an open-source software platform used to write, compile, and upload code to microcontrollers. It provides a simple interface and supports various boards including ESP32, making it ideal for developing and testing embedded applications in this project.

2. **Embedded C / Arduino Programming**

   Embedded C and Arduino programming languages are used to develop the control logic of the system. They help in interfacing sensors, controlling motors, processing input data, and executing real-time operations required for landmine detection.

3. **ESP32 Board Support Package (BSP)**

   The ESP32 BSP is required to enable Arduino IDE to recognize and program the ESP32 microcontroller. It includes necessary drivers, libraries, and configuration files to utilize features like Wi-Fi, Bluetooth, GPIO control, and camera interfacing.

4. **Wi-Fi Communication Interface (Web/Mobile Dashboard)**

   A Wi-Fi-based interface allows real-time monitoring and control of the system through a web page or mobile application. It helps in displaying sensor data, GPS location, and live camera feed, enabling remote operation and improving safety by keeping users away from hazardous zones.

PROJECT COST ESTIMATION

Sr.No	Name	Quantity	Cost
1.	Esp 32	1	500
2.	Current Sensor	1	70
3.	Voltage Sensor	1	100
4.	Temperature Sensor	1	120
5.	Relay Module	1	70
	TOTAL		860

IMPLEMENTATION

The implementation of the EV charger monitoring system begins with the integration of hardware components centered around the ESP32 microcontroller. Various sensors, including voltage, current, and temperature sensors, are interfaced with the ESP32 to collect real-time data from the charging system. Proper pin configuration and stable power supply are ensured for accurate operation of all components.

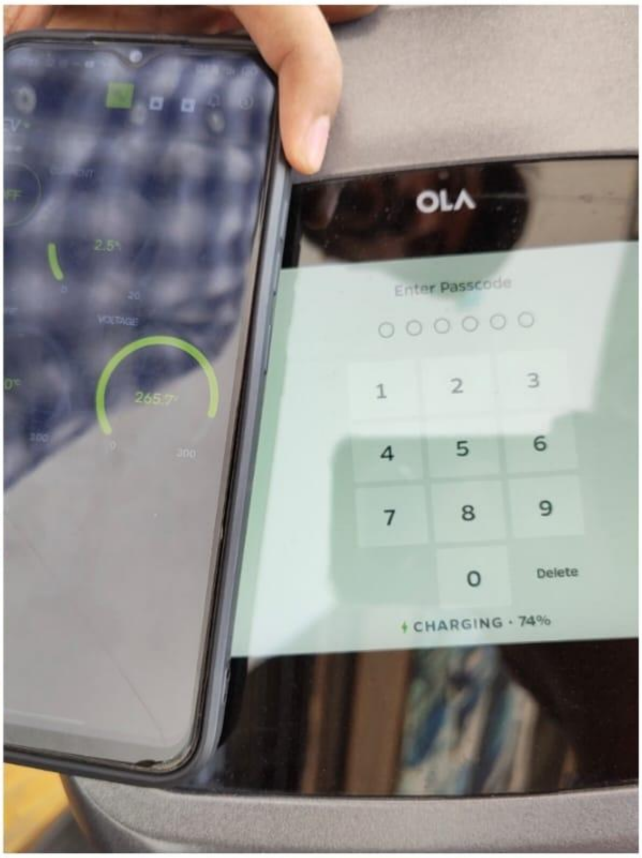
The AC input supply is connected to the EV charger through a relay module, which acts as a control switch. The relay is governed by the ESP32, allowing the system to automatically disconnect the power supply whenever unsafe conditions are detected. This provides an added layer of protection during the charging process.

The sensors continuously measure electrical parameters such as voltage, current, and temperature. These values are transmitted to the ESP32, where they are processed to calculate important parameters like power and energy consumption. Threshold values are predefined in the system to identify abnormal operating conditions.

For user interaction, the processed data is displayed on the serial monitor or an optional LCD display, providing real-time feedback. Additionally, the ESP32 transmits the data to an IoT platform such as Blynk, enabling remote monitoring and visualization through a mobile interface.

The overall system is tested under different operating conditions to verify its performance. Fault scenarios such as overcurrent and overheating are simulated, and the system successfully responds by triggering the relay to cut off the power supply.

This demonstrates the effectiveness of the system in ensuring safe and reliable EV charging.


OLA
Enter Passcode
1 2 3
4 5 6
7 8 9
0 Delete
CHARGING · 74%

RESULT

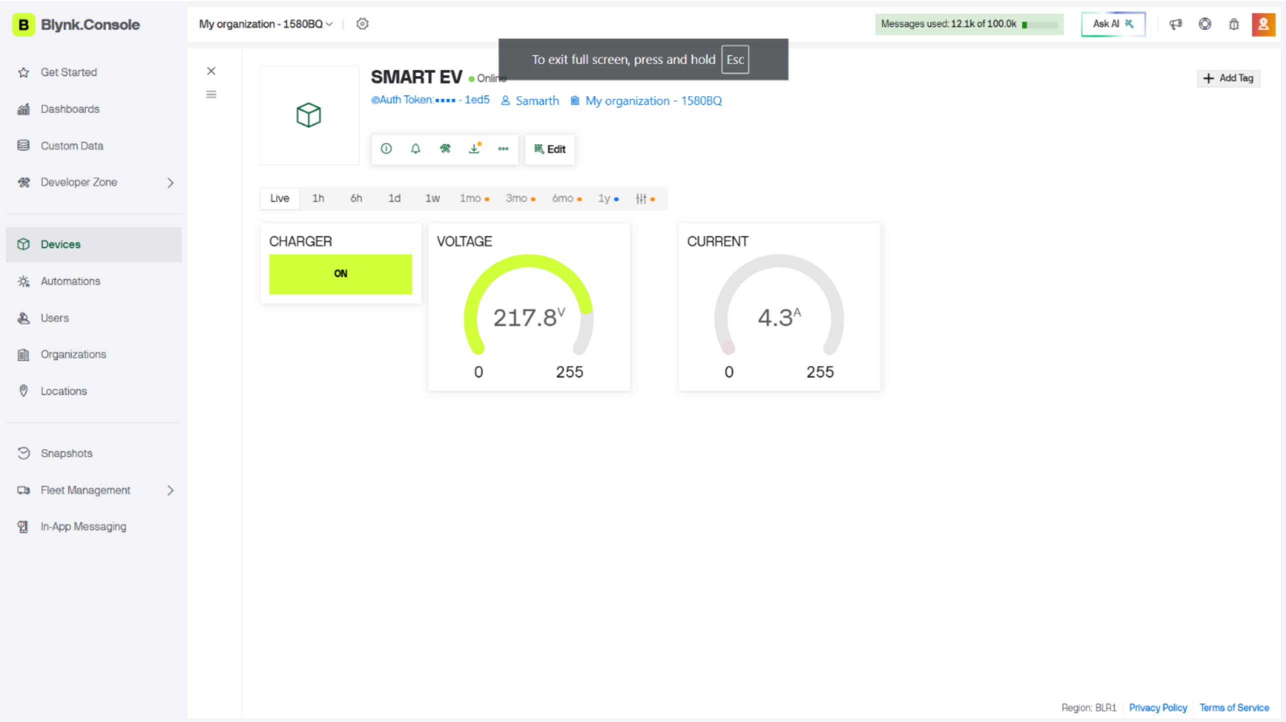
The developed EV charger monitoring system successfully acquired real-time data from voltage, current, and temperature sensors during operation. The ESP32 processed these inputs reliably, ensuring continuous monitoring without significant delay or data loss.

The system was able to compute electrical parameters such as power and energy consumption based on the measured values. The calculated results were consistent with expected behavior under different load conditions, demonstrating acceptable accuracy of the sensing and processing unit.

The protection mechanism was effectively validated by introducing abnormal conditions. When predefined thresholds for current, voltage, or temperature were exceeded, the relay module responded promptly by disconnecting the power supply, ensuring system safety.

The integration with the IoT platform, such as Blynk, enabled remote monitoring of parameters. However, minor inconsistencies were observed in real-time data updates due to network and calibration limitations.

Overall, the system demonstrated reliable performance in monitoring, control, and protection. While certain limitations were identified, the prototype effectively validates the concept of a smart EV charger monitoring system for practical applications.



FEATURES

Advantages

▢ **Enhanced Safety**
Reduces risk of electrical hazards by detecting abnormal conditions like overcurrent, overheating, and voltage fluctuations, and automatically cutting off power.

▢ **Real-Time Monitoring**
Continuously tracks voltage, current, temperature, and power, providing instant system status to the user.

▢ **Automatic Protection System**
The relay-based mechanism ensures immediate disconnection during unsafe conditions, preventing damage to equipment.

▢ **Remote Accessibility**
With IoT integration, users can monitor charging parameters from anywhere using mobile or web applications.

▢ **Energy Tracking Capability**
Calculates power and energy consumption, helping users analyze usage and estimate charging costs.

▢ **Low-Cost Implementation**
Uses affordable components like ESP32 and basic sensors, making it suitable for small-scale and home setups

Disadvantages

▢ **Sensor Accuracy Limitations**
Low-cost sensors may introduce measurement errors if not properly calibrated.

▢ **Network Dependency**
IoT functionality depends on stable internet connectivity, which may affect real-time monitoring.

▢ **Limited Industrial Robustness**
Prototype design may not meet industrial standards for high-power commercial charging stations.

▢ **Calibration Requirement**
Frequent calibration is needed to maintain accuracy of voltage, current, and temperature readings.

▢ **Basic Fault Detection**
The system detects threshold-based faults but may not identify complex or predictive failure conditions.

Applications

▢ **Home EV Charging Systems**

The system can be deployed in residential setups to monitor charging conditions in real time. It helps users detect overheating, overcurrent, or abnormal voltage without manual supervision, ensuring **safe overnight charging** and reducing the risk of electrical hazards.

▢ **Apartment & Society Charging Points**

In shared residential spaces, multiple EVs may be charged simultaneously. This system can be used to monitor individual charging points, track energy consumption, and prevent overloading of the electrical infrastructure, improving **load management and safety**.

▢ **Small Commercial Charging Stations**

For small businesses or parking facilities, the system provides **basic monitoring and protection** without expensive infrastructure. Operators can track usage, identify faults, and ensure uninterrupted service for users.

▢ **Energy Monitoring & Cost Analysis**

The system enables tracking of power and energy consumption during charging sessions. This data can be used to estimate electricity costs, analyze usage patterns, and support **billing systems or energy optimization strategies**.

▢ **EV Service & Maintenance Centers**

During servicing, technicians can use the system to monitor charging behavior and detect irregularities such as excessive current draw or temperature rise. This helps in **diagnosing faults in chargers or vehicle batteries**.

▢ **Smart Grid & Future Integration**

The system can act as a base model for integration with smart grid technologies, where charging loads are managed dynamically. It can contribute to **load balancing, peak demand control, and efficient energy distribution**.

CHALLENGES FACED

☐ **Sensor Accuracy and Calibration Issues**

Voltage, current, and temperature sensors showed inconsistent readings due to calibration errors and environmental noise. Achieving accurate and stable measurements required careful tuning and testing.

Solution:
Sensor values were calibrated by comparing readings with standard measuring instruments like a multimeter. Multiple test readings and software correction factors were used to improve accuracy and stability.

☐ **Temperature Measurement Inaccuracy**

The temperature sensor did not provide correct values on the ESP32 serial monitor initially, mainly due to improper wiring and placement, affecting reliable overheating detection.

Solution:
Proper sensor wiring was verified and corrected according to the datasheet. The sensor placement was improved near the charger heating area to obtain more accurate temperature readings.

☐ **IoT Data Synchronization Issues**

Voltage and current values were not updating correctly on the IoT dashboard due to data mapping errors and network delays, impacting real-time monitoring performance.

Solution:
Correct virtual pin mapping and optimized data transmission intervals were implemented in the Blynk IoT platform. Stable WiFi connectivity also improved real-time synchronization.

☐ **Signal Noise and Fluctuations**

Analog sensors introduced noise and fluctuations in readings, making it difficult to obtain stable outputs without implementing filtering techniques.

Solution:
Software filtering techniques such as averaging multiple sensor readings were applied to reduce fluctuations and obtain stable output values.

☐ **Relay Control Stability**

Ensuring proper relay operation was challenging, as incorrect threshold settings led to false triggering or delayed response during fault conditions.

Solution:
Threshold values for voltage and temperature were carefully tested and optimized. Proper relay control logic with delay handling was implemented to avoid false triggering.

☐ **Power Supply and Hardware Issues**

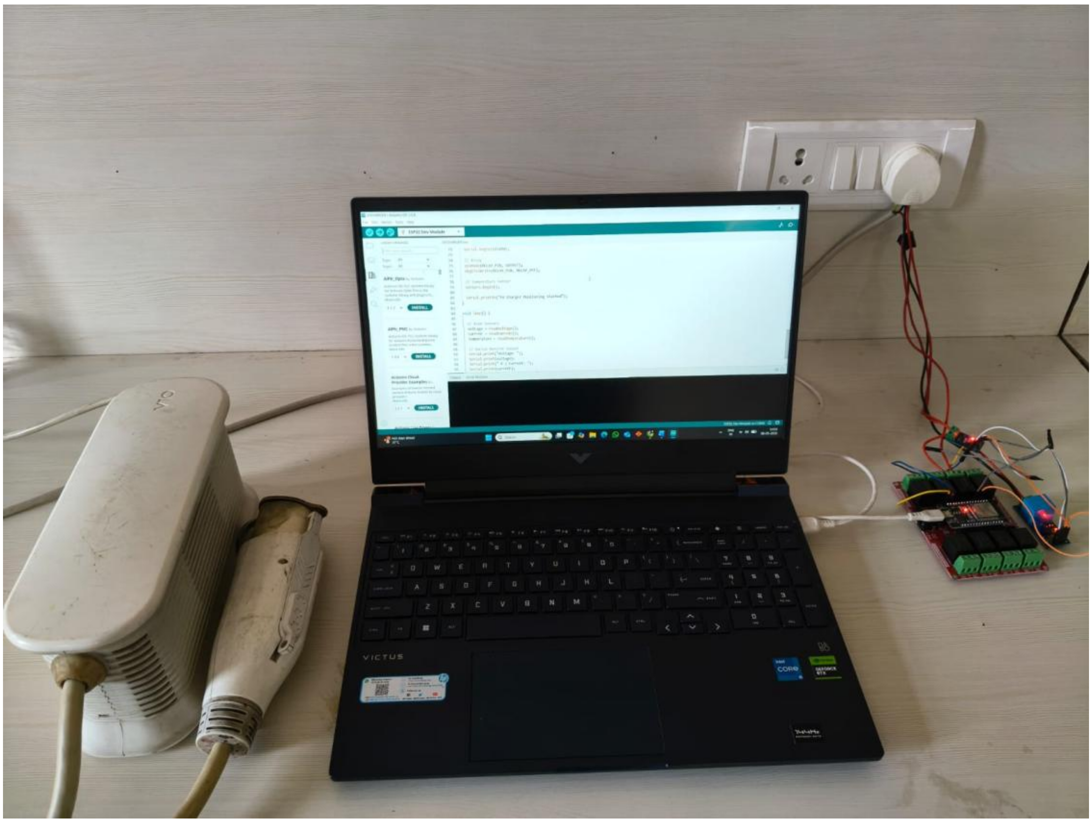
Maintaining a stable power supply for ESP32 and sensors was difficult during initial setup, and loose connections affected system performance.

Solution:
A regulated power supply and secure wiring connections were used to maintain stable operation. Hardware connections were checked regularly during testing.

☐ **System Integration Complexity**

Integrating multiple components such as sensors, ESP32, relay module, and IoT platform required proper synchronization, making the overall system design technically challenging.

Solution:
The system was developed and tested module-by-module before final integration. Proper synchronization between hardware and software components ensured smooth overall operation.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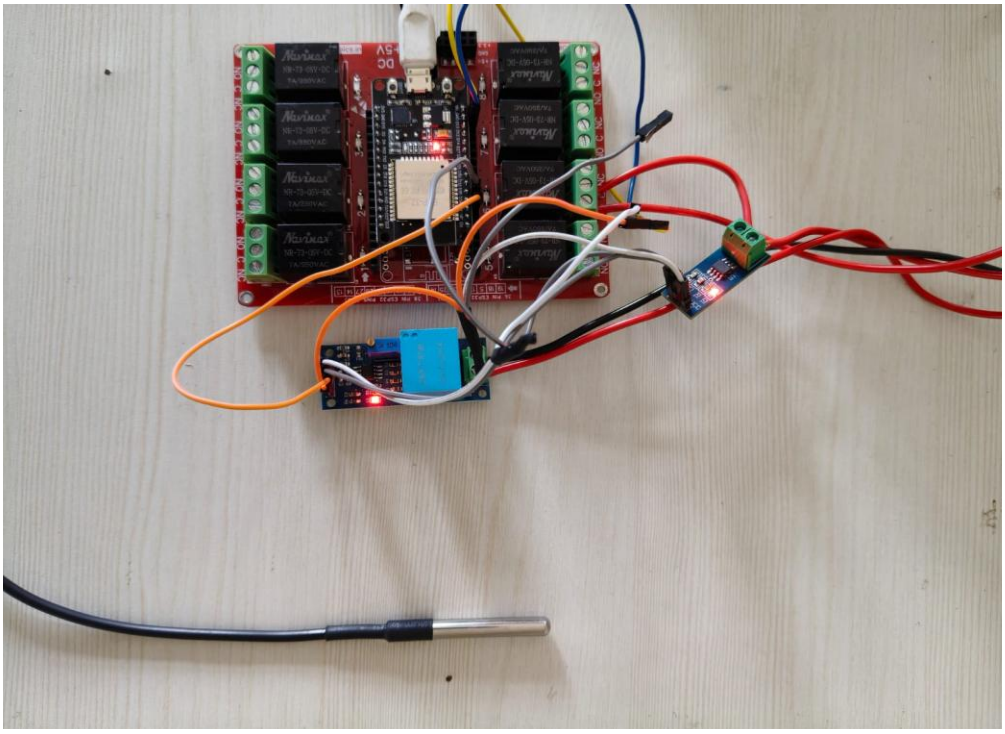

CONCLUSION

The EV charger monitoring system was successfully designed and implemented using the ESP32 microcontroller along with voltage, current, and temperature sensors. The system was capable of continuously acquiring and processing real-time data, which demonstrates the effective application of embedded systems in modern electric vehicle charging solutions. The integration of multiple sensors enabled comprehensive monitoring of critical electrical and thermal parameters during the charging process.

The monitoring functionality provided accurate and reliable observation of voltage, current, and temperature values. Based on these inputs, the system was able to calculate power consumption, offering useful insights into the performance and efficiency of the charging system. This helps in understanding energy usage patterns and improving overall energy management.

To ensure safety, a protection mechanism was incorporated using a relay module. The system was programmed to automatically disconnect the power supply whenever abnormal conditions such as overvoltage, overcurrent, or excessive temperature were detected. This feature significantly enhances the safety of the charging process by preventing potential hazards like overheating, equipment damage, or electrical faults.

The project also demonstrates the importance of real-time monitoring and automated control in improving the reliability of EV charging systems. By combining sensing, processing, and control in a single platform, the system provides an efficient and user-friendly solution for smart energy management.

Overall, the proposed system offers a low-cost, scalable, and effective approach to enhancing the safety and efficiency of EV chargers. Although minor challenges were encountered during implementation, such as sensor calibration and signal accuracy, the project successfully validates the concept of smart monitoring. With further improvements such as IoT integration, mobile app control, and advanced analytics, this system has strong potential for real-world deployment and future development.

FUTURE SCOPE

The system can be further enhanced by developing a dedicated mobile or web-based application that provides advanced data visualization, real-time alerts, and remote control functionality. This would significantly improve user interaction and convenience, allowing users to monitor and control the EV charger from anywhere. Integration with IoT platforms would also enable cloud-based data storage and analysis.

Another important improvement is the addition of a cost estimation and billing system based on energy consumption. By calculating the total energy used during charging sessions, the system can generate accurate billing information. This feature would be highly beneficial for public charging stations and shared infrastructure, ensuring transparency and enabling automated payment systems.

The project can also be upgraded by incorporating predictive fault detection techniques using basic data analytics or machine learning algorithms. By analyzing historical sensor data, the system can identify abnormal patterns and predict potential failures in advance. This proactive approach would enhance reliability and reduce maintenance costs by preventing faults before they occur.

In future developments, the system can be expanded to support multiple EV charging points simultaneously. This scalability would make it suitable for large-scale applications such as commercial charging stations, apartment complexes, and parking facilities. Efficient load distribution and centralized monitoring would be key features in such implementations.

Additionally, integrating the system with renewable energy sources such as solar panels and smart grid technologies can further improve its sustainability. This would allow optimized energy usage, reduced dependency on conventional power sources, and better load management. Such integration aligns with the growing demand for green energy solutions in modern EV infrastructure.

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